

AE 4342 Space System Design
Final Report

**PONYO: Polar Orbiting and Yielding Observations
CubeSat**

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Spring Semester 2023

This report was written in L^AT_EX.

1 Introduction

The Polar Orbiting and Yielding Observations (PONYO) CubeSat is a water-propelled CubeSat for the pursuit of planetary exploration and research in water-propulsion systems, the efficacy of water radiation shielding, and lunar mapping. Our clientele, Georgia Tech’s research lab and NASA ,for the Small, Innovative Missions for Planetary Exploration (SIMPLEx) challenge, have provided the challenge of combining planetary exploration and research of water in a space environment into a small CubeSat.

PONYO is a 6U CubeSat that utilizes a Magnetohydrodynamic drive for its water-based propulsion, reflectance spectroscopy for the mapping of lunar polar caps in the search for evidence of water while in a near-rectilinear halo orbit (NRHO), and Geiger counters, both on the propulsion system’s water tank and away from any water-filled systems, to obtain further data on water’s effectiveness as radiation shielding. These innovative architecture selections were based on numerous literature reviews to best justify and meet the needs presented.

While staying within the requirements of the SIMPLEx competition, as well as being around 1 million dollars under the 5.6 million dollar budget, PONYO is scheduled to launch with the SLS in 2028 and close out the mission within a year of launch. This is the overall cost budget for the mission which utilizes an analogy estimation and a simple linear regression approach to the remaining phases. See Appendix 4 for the cost budget. To compensate for budget weaknesses, we have included a more conservative estimate and priced for the most expensive scenarios to give us a large enough buffer for future complications.

Since PONYO’s mission is an experimental run with the express intent of testing and demonstration, there is a realistic chance of failure due to this experimental nature. However, all instrumentation used is at or above a TRL (Technology Readiness Level) of 6, which mitigates some risk of instrumentation failure. Fig. 1 further displays an updated risk matrix.

		PONYO Risk Matrix				
LIKELIHOOD	5					
	4					
	3		Loss of Comms			
	2	Cost Growth	Radiation Shielding Failure	Propulsion Failure	Power Failure	
	1	Quality Control Insufficient	Observation Instrumentation Failure	Trade-off Design Issues		
		1	2	3	4	5
		CONSEQUENCES				

Figure 1: Risk Matrix

All of these risks together means PONYO falls within a class D risk, but whether or not this mission completes all tasks, such as entering the desired science orbit of NRHO (Near-Rectilinear Halo Orbit), any and all data obtained and sent back is seen as a success and will be used to further research on planetary exploration and potential uses of water in space missions.

The Concept of Operations is shown in Fig 2. This illustrates the main 8 phases that PONYO will undergo throughout the mission’s lifetime.

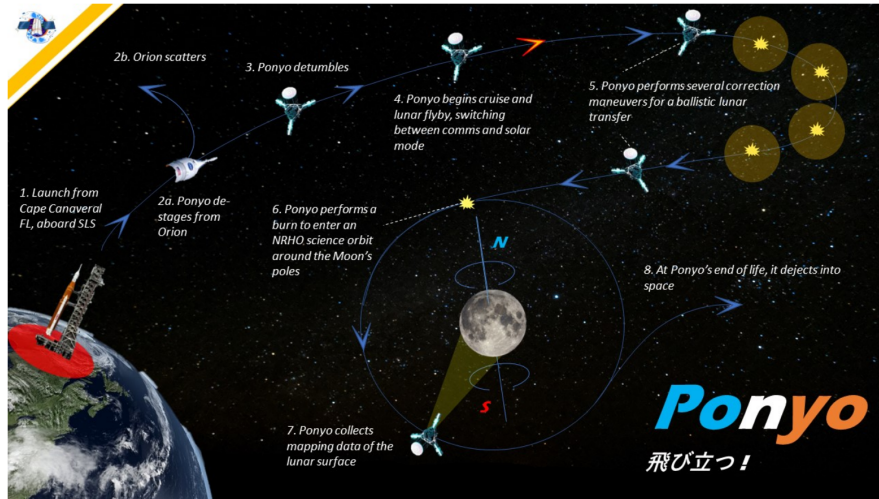


Figure 2: PONYO Concept of Operations

2 System and Configuration

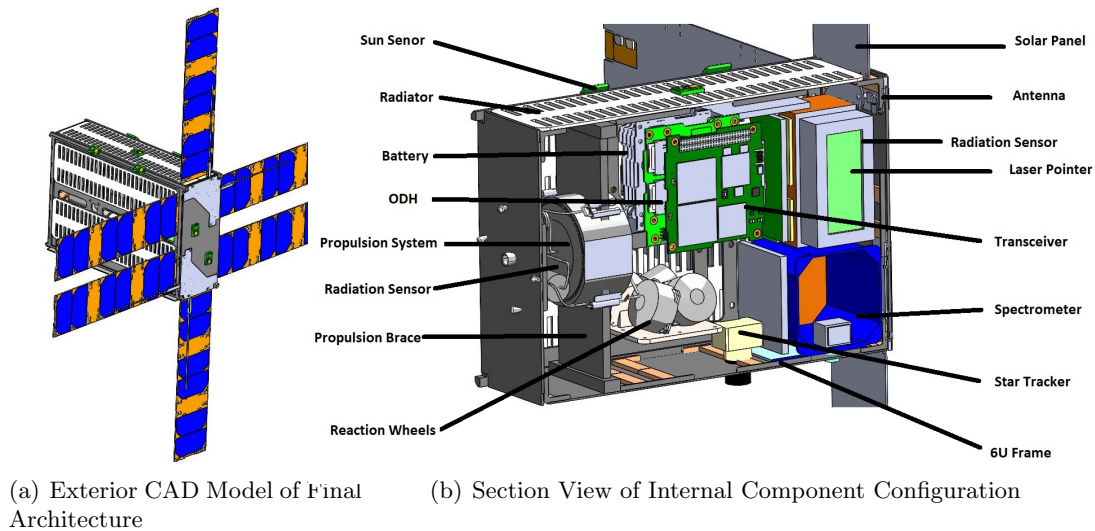


Figure 3: Final Architecture System and Configuration

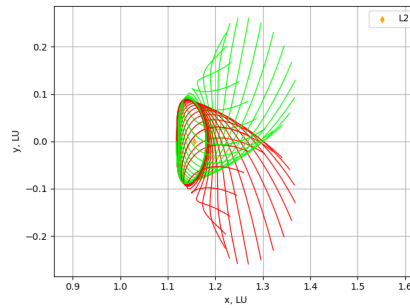
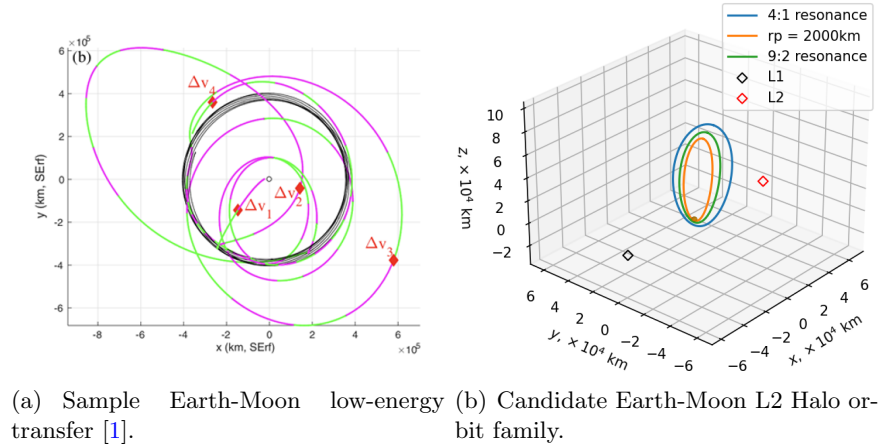
In this section, the optimal configuration of PONYO is explored and explained with the configuration guidelines covered in lecture. As presented in Fig. 3(a) depicting the ideal exterior configuration of PONYO while in orbit, the six exterior 3U solar panels are shown deployed and attached to the front face of the CubeSat for maximum power generation. Immediately upon deployment from the launch vehicle (LV), spring-loaded mechanisms will activate, releasing the solar panels from their initial, closed position to the open position seen in the figure. In Fig. 3(b), all of the internal components are labeled and their positions in relation to one another are shown. The payload, consisting of the Laser Pointer and Spectrometer fills up 2U of volume, and to avoid interference from the propulsion system, was placed at the front of the CubeSat; the Transceiver, On Board Data Handling, Battery, and Star Tracker were also clustered around payload to minimize cabling. The propulsion system and reaction wheels were the heaviest components in the configuration, so their placement was as close as possible to the center of mass to minimize torque and account for propellant loss

during the mission, where the displacement of the center of mass had to be minimized. The exterior radiators acted as external paneling for the 6U structure to enclose all of the internal components, passive thermal control with their function to retract and expand, gaps to release heat, and mounting plates for the fourteen sun sensors located around the external frame. To fulfill the radiation shielding mission, two Geiger Counters are placed within PONYO, one mounted to the water tank to be the variable data and another with the payload at the front end of the CubeSat to be the control data. The difference in radiation measurements at these two locations will help determine how effective water is in the shielding of radiation.

3 Mission Design and Analysis

In this section, the mission design of PONYO and trajectory analysis are conducted. There are three stages of trajectories that need to be designed in the mission: Earth-Moon transfer, cis-lunar orbit, and end-of-life maneuver. The following subsection elaborates on each trajectory with the corresponding trade space of ΔV and time of flight (ToF). Additionally, the science mission around the south pole is briefly discussed.

3.1 Earth-Moon Transfer



(c) Stable (green) and unstable (red) manifolds in an L2 Halo orbit..

Figure 4: PONYO trajectory design.

Leveraging the unique dynamics of the Sun-Earth-Moon system is a prerequisite for the low- ΔV transfer trajectory design to the cis-lunar domain. A design method using bi-circular restricted four-body problem (BCR4BP) is performed in the EQUULEUS trajectory design,

and a sample solution is presented in Fig. 4(a), where $\text{ToF} = 207$ days and $\Delta V = 13.7$ m/s. The trade space of the transfer trajectory is $\text{ToF} = 100\text{-}200$ days, and $\Delta V = 10\text{-}20$ m/s.

3.2 Science Orbit

North L2 9:2 resonance near-rectilinear Halo orbit (NRHO) is chosen for the target orbit for PONYO. This orbit has several advantages such as no Earth eclipse due to its unique resonance, high stability, and (relatively) low periapsis radius ($r_p \simeq 4000\text{km}$), which aligns with our scientific objective. Additionally, as an extended mission, $r_p = 2000\text{km}$ NRHO is also considered. This orbit has a minimum altitude of 250 km, which would give us a significant scientific output in compensation for additional ΔV consumption. These trajectories are shown in Fig. 4(b). Based on the EQUULEUS mission, the station-keeping cost is estimated as 10 m/s throughout the mission.

3.3 End of Life

Since the chosen NRHO's stability index is larger than 1, the weak stability boundary effect in the circular-restricted three-body problem (CR3BP) can be considered. This will enable us to leverage the unstable invariant manifold (Fig. 4(c)) and deorbit the spacecraft from the cis-lunar region with effectively no cost.

3.4 Science Payload

PONYO executes a spectroscopic observation campaign at the south pole region after it is inserted into the NRHO. Active reflectance spectroscopy is chosen as a scientific payload, where the laser pointers have four distinct wavelengths of 1, 1.5, 1.8, and 2 μm . For each laser pointer's output, 99% of them is expected to propagate in 8 mrad, and the receiver's field of view is set to 10 mrad. With this assumption, we can expect the resolution of 15 km in 9:2 resonance NRHO, and 5km in $r_p = 2000$ km NRHO, which can sufficiently capture the surface property of the lunar south pole region. Even though there are constraints in the effective focal distance, COTS Indium gallium arsenide (InGaAs) with a diameter of $D = 0.25$ mm is available, which satisfies this mission requirement. The boresight alignment error is expected to be 1 mrad.

4 Structure

A 6U CubeSat structure was selected to encapsulate all of the internal components necessary for this mission, as shown in Fig. 5. The standard size for a CubeSat unit is a 10 cm cube measured externally, although these measurements vary by a few centimeters for each manufacturer. The structure selected for the mission was narrowed down to ISISpace's normal 6U structure sized at 10 cm $\times$ 22.6 cm $\times$ 34.05 cm. The overall mass of the frame and its internal structural support for the components, primarily the propulsion system, is estimated at 1.15 kg. The entire frame and sub-structures are made of the industry's standard material, Aluminum 6061, for its availability, strength, and low weight. The redundancy measures taken into account are shown with the propulsion sub-structure.

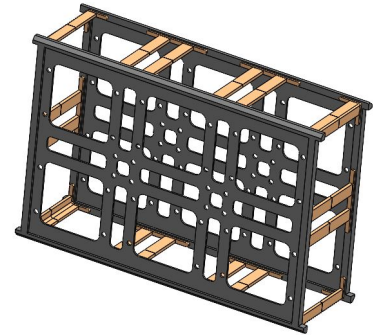


Figure 5: 6U CAD Structure

The Propulsion system will be fixed to the back panel along with two supports fitting to the water tank on either side to keep from shifting during PONYO's mission. Bottom-Up mass and volume budgets were created for all components to ensure they would fit within the volume parameters and estimate the total mass of the CubeSat. These budgets are shown in Appendix A.4 with Fig. 13, which indicates the volume parameter is met and the total mass was estimated at 11.23 kg. The only load to consider on the structure is thermal loading from the temperature oscillations. The temperature range Aluminum 6061 can endure and not deform is approximately 150 °C, while the highest environmental temperature experienced will be 42.91°C. This indicates that the structure is well within the range to operate without facing plastic deformation.

5 Propulsion

5.1 Configuration of the propulsion system

The propulsion system consists of three main components: a main nozzle attached to a combustion chamber, which provides the necessary ΔV for orbital maneuvers, at least 4 small, cold gas thrusters for desaturating the reaction wheels, and the Magnetohydrodynamic drive (MHD) drive [2].

The optimal operating pressure for the propulsion system is discussed in A.1, and will be set at 10 bar. A capacitor sensor is used to open the valves when the gas produced in the limited volume of the MHD drive increases the pressure until it reaches the desired value. Additionally, a flashback arrestor prevents reverse flow from the combustion chamber, which could lead to an explosion in the MHD drive. All valves are duplicated for redundancy, except for the ones leading to the combustion chamber, as the mass flow rate toward it is higher. A diagram depicting the different elements of the propulsion system is shown in Fig. 6.

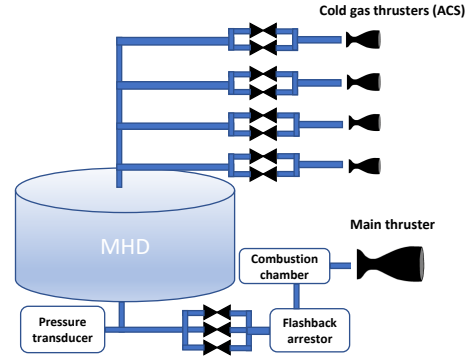


Figure 6: Different components of the propulsion system and their connect with each other.

5.1.1 Desaturation system configuration

The desaturation system of CubeSats shall be capable of desaturating the reaction wheels in all three axes (X, Y, and Z). The thrusters are strategically positioned, as illustrated in Fig. 7(a), to achieve this purpose.

5.2 Sizing

5.2.1 Nozzle and Combustion Chamber Sizing

For the nozzle sizing, the objective is to increase the specific impulse, I_{sp} . This can be achieved by maximizing the thrust coefficient, C_F , which requires a high nozzle expansion ratio, ϵ . However, as shown in Fig. 3-7 of Sutton [3], the variation of C_F with ϵ is asymptotic. Additionally, to avoid excessively long burn times, a minimum mass flow rate must be maintained through the nozzle, and an extremely high value of ϵ would result in an absurdly long

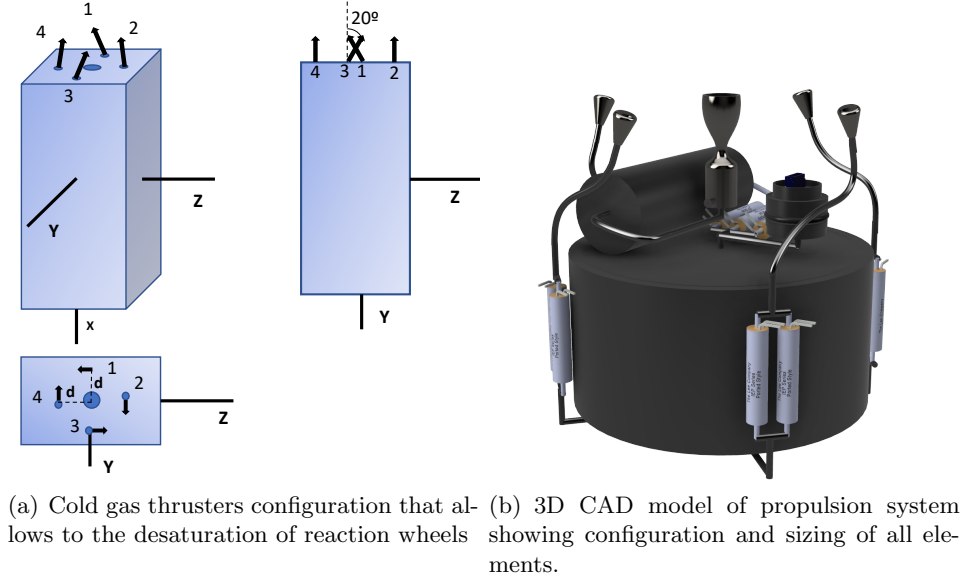


Figure 7: Configuration and 3D model of propulsion system.

Table 1: Summary of dimensions and expected performance for the propulsion system

	ϵ	D_t (mm)	D_e (mm)	L_n (mm)	$\dot{m}$ (g/s)	I_{sp} (s)
<i>Cold gas thruster</i>	200	0.225	3.2	5.5	0.097	119.71
Main thruster	62	1.27	9.38	17.5	0.5	440.28

nozzle. Therefore, a balance must be struck between the nozzle size and the thrust coefficient. The values shown in Table 1 were obtained by finding this balance, and the calculations and discussions behind these values can be found in Section A.2.

For the sizing of the combustion chamber, the characteristic length L^* can be obtained from Table 5.6 in [4], which gives $L^* \in [56, 71]$ cm. Here, the high value is used, resulting in a characteristic length of $L^* = 71$ cm. Using this value and the throat area A_t , the volume of the combustion chamber can be found as $V_{cc} = L^* A_t = 0.90$ cm³. Using the correlation explained in Section A.2.2, the area $A_{cc} = 0.425$ cm², diameter $D_{cc} = 0.681$ cm, and length $L_{cc} = 2.46$ cm of the combustion chamber are obtained.

5.2.2 Cold Gas Thrusters Sizing

The purpose of the thrusters is to desaturate the reaction wheels [5] on the CubeSat. Reaction wheels spin to maintain the CubeSat's stability, but over time, they may reach their maximum velocity and need to be slowed down. To accomplish this, an external torque is applied to the CubeSat by the thrusters and the reaction wheels compensate for the external torque by reducing their rotation speed. For the reaction wheels used on the CubeSat, the maximum momentum stored is 37 mN·m·s. An $\epsilon = 200$ shows to balance the performance and size of the cold gas thrusters, as more deeply explained in A.3 and shown in Table 1. These dimensions allow desaturation of rotation wheels in 10 s.

6 TMTC & ODH

Throughout the course of the mission, PONYO will have to relay the data that it takes back to Earth for evaluation. Once detumbling is completed after exiting the SLS, the antenna will

expand as PONYO makes its way toward the moon. Then, after an NRHO has been entered, periods of telecommunication and charging will alternate throughout the seven day period. When near the lunar south pole, though, PONYO will enter science mode meaning that the laser measurements and positions will be taken and saved to the on board computer to be transmitted to Earth once out of range of the south pole.

6.1 Transmission

The measurements taken during the science mode will require the most storage and therefore require the highest downlink rate. Laser and position measurements will be taken over 5 minutes totaling around $6 \cdot 10^7$ samples totalling 1.29 Gb to be transmitted during the next orbit. Two thirds of the time will be spent charging, which allows for a maximum downlink rate of 7.47 kb/s. To achieve this goal, an omnidirectional, dipole antenna will be utilized with a maximum gain of 3 dB (GOMspace 6U antenna system). This gain drops too drastically beyond 30 degrees from the optimal angle to achieve accurate transmissions. The antenna transmits in the UHF frequency band at 400 MHz and receives in the VHF frequency band at 161 MHz. All attitude and navigation will be done autonomously using sun sensors and star trackers. Uplink is reserved for emergency attitude/trajectory adjustments if the autonomous systems malfunction. This requires a budget around 5 kb/s. The LeafSpace Ground Station Network offers their dishes to be rented commercially for lunar missions. These dishes have the datarate, frequency range, and gain capabilities required for the scope of this mission. A cyclic redundancy check algorithm will be implemented and monitor the transmissions to ensure that data is being received in the same state that it is transmitted. The antenna requires 2.3 W while deploying, but only needs 0.03 W afterwards and has a mass of 0.09 kg. The ISISpace VHF/UHF duplex transceiver has a power consumption of 4 W and a mass of 0.075 kg.

6.2 On-Board Computing

As mentioned previously, the maximum data being stored at any one time will be 1.29 Gb following our pass-by of the lunar south pole. This is addressed by having 2 2 Gb SD cards in the on-board computer (OBC). Additionally, to reduce the amount of data being transmitted, small calculations will be done on board. This primarily includes calculating the lunar location for each laser measurement given PONYO's orientation and position. The ISISpace on-board computer (iOBC), which was selected for this mission, has a 32 bit ARM9 processor with a clock speed of 400 MHz. Even if the calculations take 2-3 clock cycles, this processor is more than capable of completing these calculations in real time. To achieve proper connections between all necessary modules, an I^2C bus will be used. Double connections between components will ensure that in the event of one failing, the second connections prevails. The clock speed of this bus is 1 MHz. The laser and star tracker require the largest movement of data, and the clock speed of this bus is more than enough to keep up with the influx of data. Overall, the OBC requires a maximum power of 0.8 W and has a mass of 0.176 kg.

7 Power

The power for PONYO will be supplied via six Blue Canyon Technologies deploy-able, rigid 3U BCT Solar Arrays and one AAC Clyde Space Optimus-40 lithium polymer secondary battery. Using six 3U BCT solar arrays ensured that there was a generous margin between power provided and power required during each of our 7 modes. Peak power output was determined by the solar arrays' total PEOL which was determined to be about 205.7W based

on the duration of the mission and the degradation rate of GaAs. This ensures that the power system would easily be able to supply the most expensive power request of 127W with the help of a secondary battery. Figure 8 below provides an accurate estimation of power supplied to power required for each mode of the mission showing that all systems are capable of being powered for each mode.

Subsystems	Power Provided, (We)							
	Specific Peak Power Output	Cruise Phase Modes			Science Phase Modes			End of Life
		Mode 1 Launch	Mode 2 Cruise	Mode 3 NRHO Insertion	Mode 4 Nominal	Mode 5 Data Collection	Mode 6 Data Transmission	Mode 7 Eject from NRHO into Space
4.0 Power Provided	245	30.75	184.5	205	205	132.25	123	164
4.1 Deployable Solar Panels (x6)	205	30.75	184.5	205	205	92.25	123	164
4.2 Secondary Battery	40	0	0	0	0	40	0	0
Power Required	*	20.95	53.321	111.051	24.051	126.98	24.051	22.021

Figure 8: Power Provided vs. Power Requested for Each Mode

During the launch, the solar panels will be stowed to conserve space and only when PONYO ejects from the SLS and detumbles will the spring-loaded hinges be released and the solar panels deployed. It can be assumed that solar eclipsing will have little to no effect due to choosing the specified NRHO. The vast majority of the science orbit will be sunlit having only an 18 minute eclipse window in 1-2 of the orbits representing roughly 0.016 % of the science phase. The only time where the solar panels would be insufficient on their own would be during the data collection mode where the charged secondary battery would be in use to supply the extra power needed holding a total charge capacity of 40 Wh. For redundancy, all powered systems will have double wire connections to the power supplies to ensure that if one connection breaks during the mission, the other will still be able to provide power. In addition, the wires themselves will be wrapped in Kapton tape to ensure they are properly insulated as a form of thermal protection. Lastly having six 3U solar panels not only provides PONYO with a sufficient power margin but also protects against mission failure if one panel should fail to deploy in the detumbling phase of the mission. The mission could be carried out with only five generating optimal power.

8 ADCS

The attitude determination control system (ADCS) ensures that PONYO can maintain an axis, accomplish pointing requirements, self-stabilize, and maneuver in an efficient manner. Throughout the mission's duration, the CubeSat must be able to generate power with its solar panels, collect data from the payload, and relay data to the Earth, each requiring the ADCS for accurate pointing.

Seven control modes ensure that component pointing needs are met at different mission phases: acquisition, charging, data transfer, orbital maneuvers, science, slew, and contingency. These seven modes can be seen in Fig. 9.

PONYO will intermittently switch between the charging and data transfer mode in order to ensure optimal sun pointing (0° incidence angle) and optimal antenna pointing (90° to the ground station). Two hours spent in charging is able to support one hour of data transfer even at the worst case scenario of a 90° incidence angle.

Multiple sensors are used in order to ensure pointing requirements are met. Coarse sun sensors with a field of view, FOV, of 85° are placed on each face of the Cubesat, except the face on which thrusters are placed. Furthermore, pyramidal platforms on each corner allow

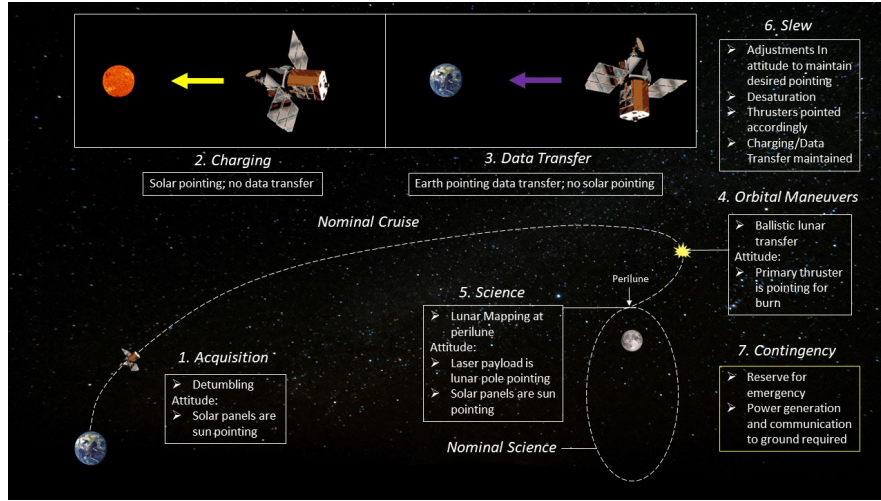


Figure 9: Diagram of ADCS modes and their corresponding relevance to the mission timeline

for the placement of two sun sensors oriented 45 degrees from the face, determining a 1D reference direction vector[6]. Overall, 14 sun sensors ensure a more than sufficient FOV of the sun and allow for proper sensing even if half are inoperable. Another sensor used is an IMU which encompasses three MEMS gyros, three accelerometers, and three inclinometers. The IMU can be placed anywhere within the Cubesat especially due to the low temperature bias, $2^\circ/\text{rms}$, of its components. The last sensor employed is a star tracker, which along with either one of the aforementioned sensors establishes a redundant three axis attitude control system.

During acquisition only the sun sensors are used for pointing, whereas the IMU is employed to measure the angular velocity of the Cubesat. Using these inputs a B-Dot control algorithm can be employed to detumble [7]. The solar panel and antenna pointing will use a sun sensor and star tracker. Thruster pointing will utilize IMUs and star trackers while the desaturation will only use IMU. Finally the payload pointing will need to use IMUs and sun sensors to help maintain the CubeSat in the correct direction and orbit. Such sensors were necessary due to the harsh environments that the CubeSat will endure and the torque encountered from the sun, earth, and moon [8].

The IMUs, sun sensors, and star tracker will guide the attitude control system, which is comprised of four reaction wheels and four thrusters.

Table 2: Quantification of external and internal torques

External Torque	Value (N-m)	Internal Torque	Value (N-m)
Earth gravity gradient	8.05E-07	Propellant Sloshing	baffle mitigates
Moon gravity gradient (perilune)	3.93E-04	Reaction wheel friction	9.6E-03
Moon gravity gradient (apolune)	6.04E-04		
Solar Radiation	2.25E-07		

9 Thermal Control System (TCS)

While in space, PONYO will experience a wide range of temperatures. The satellite will be subjected to high temperatures when in direct sunlight, and low temperatures when in eclipse behind the moon. To protect the internal components of the satellite, it must be equipped with a thermal control system. Generally, for a CubeSat like PONYO, an extensive

active thermal control system is not required since the volume inside the satellite is very low compared to a typical spacecraft.

To decide which thermal system to use, a lumped capacitance thermal analysis of the satellite is performed to find how different subsystems are affected by different heat sources in space. We do this by breaking up the most important subsystems into nodes, as shown in Fig. 10

Using a MATLAB code developed for the thermal analysis, we calculated the maximum and minimum temperatures that would be experienced by each node inside PONYO. As seen in Fig. 10, we account for the heat of the sun and infrared heat going into the system. We are able to assume that no heat comes via the albedo of the moon since it accounts for a nearly negligible amount of heat entering our system. This allows us to assume steady-state temperatures during PONYO's orbit, as seen in Table 3.

With the thermal analysis complete, we could choose our thermal control systems. The payload, a laser used for mapping the moon, generates a lot of heat, so the internal temperature of the satellite must be mitigated by radiating heat out. The heat inside cannot get too cold either, or the water inside the propulsion system will freeze. We also have to be careful of incoming heat, as the exterior of the satellite can get as hot as 42.91 C°. The temperature must be kept in balance to both keep the electronics at a safe operating temperature and keep the fuel a liquid. Knowing this, we decided on two passive systems. The first is AZ-93 white thermal paint. This paint has a high emissivity of 0.91, and an absorptivity of only 0.15, meaning that it excels at both radiating heat out of the satellite and reflecting any excessive incoming sunlight. The second

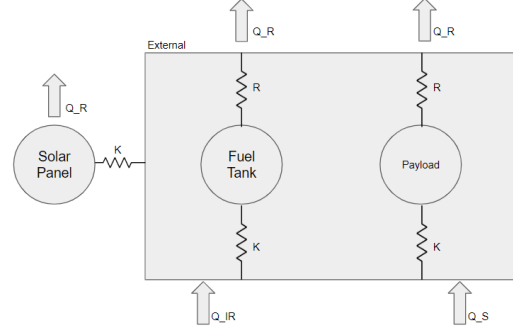


Table 3: Steady State Temperatures for PONYO in Orbit

Component	Sunlight Temp (C° [K])	Eclipse Temp (C° [K])
Exterior	42.9 (306.1)	-58.2 (207.0)
Solar Panel	26.6 (299.7)	-42.3 (230.8)
Fuel Tank	29.1 (301.3)	5.0 (278.1)
Payload	31.1 (304.3)	15.1 (288.3)

system is a passive thermal louver mounted underneath the solar panels on the sides of the satellite. These louvers work by using bimetallic springs that expand when hot, opening radiator flaps to mitigate temperature, then contract when cold, closing the flaps to conserve any extra heat we want inside. These louver systems are 3D printed, which allows them to be constructed to any specifications needed by any CubeSat. They are made of Ti64 titanium alloy for ease of printing. This system employs redundancy since each of the many flaps operates independently of one another, so if one breaks, there are many left and the system is not compromised. Another benefit is that this system operates automatically, and will mitigate heat if there is a part of the satellite that is hotter than the other, like near the payload.

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A Appendix A

A.1 Optimal operating pressure

The operating pressure chosen for a rocket engine is a critical factor that directly impacts its thrust and the available ΔV (Eq. 2-14 and 4-6 in [3]),

$$F = \dot{m} I_{sp} g_0 \quad \text{and} \quad \Delta V = I_{sp} \cdot g_0 \ln \frac{m_f}{m_i}, \quad (1)$$

due to the significant dependence of mass flow rate on combustion chamber pressure (Eq. 3-24 in [3]). However, in the current propulsion system, the mass flow rate is primarily driven by electrolysis, which produces mass at a slower rate than the mass flow rate through a thruster. The MHD drive produces around 3 grams of gas per hour which is much smaller than the

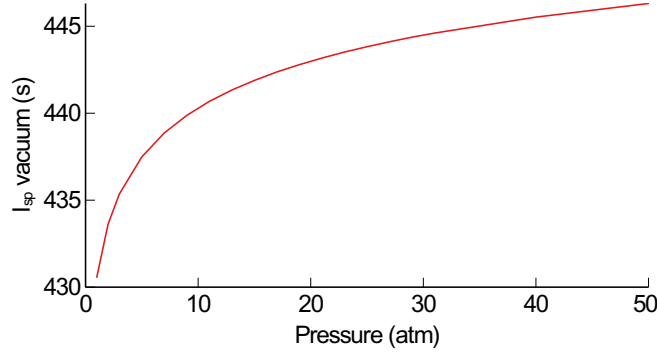


Figure 11: I_{sp} variation with operating pressure for a stoichiometric mixture of H_2 and O_2 at 300K obtained with CEA from NASA.

mass flow rate for a thruster, resulting in pulsed propulsion. The gas is produced until the desired pressure is reached, followed by a burn. Thus a higher operating pressure does not increase the mass flow rate, which is driven by electrolysis.

On the other hand, increasing the operating pressure raises the final temperature of combustion products, increasing the average exhaust velocity v_2 and the characteristic velocity c^* (Eq. 2-16 and 3-32 in [3]). This also leads to a higher specific impulse (Eq. 3-32 in [3]):

$$I_{sp} = \frac{c^* C_F}{g_0}, \quad (2)$$

as evidenced by results from NASA's Chemical Equilibrium Applications (CEA) software [9] in Fig. 11. However, beyond a certain pressure value, performance improvements become asymptotic. Moreover, higher pressure requirements for the tank and pipes make it less practical to further increase the pressure. Therefore, an operating pressure of 10 bar is chosen, providing a high specific impulse while maintaining relatively low structural requirements for the tank, this pressure is very close to the pressure chosen in the Cislunar design [10].

A.2 Main thruster sizing

A.2.1 Nozzle sizing

For the nozzle sizing, the objective is again to increase the I_{sp} . In order to do so, the C_F should be maximized by having a high nozzle expansion ratio, ϵ . It can be observed from Fig. 3-7 in Sutton [3] that the variation of C_F with ϵ is asymptotic. If a minimum mass flow rate wants to be maintained through the nozzle, and it is also desired to avoid an extremely long nozzle, this value of ϵ should not be really high. For example, by setting a $\epsilon = 10000$, and aiming to maintain a minimum mass flow rate of 1 g/s (in order to avoid excessively long burns) the nozzle throat diameter would have a minimum value of 1.7 mm (using Eq.3-24 in Sutton) and the corresponding nozzle exit diameter would be 170 mm, for a CubeSat these values are way off limit. Thus, a balance between the nozzle size and the C_F must be done.

The aim of the CubeSat propulsion system design is to save space by limiting the nozzle length to a maximum of 1.75 cm. Thus, the exit diameter can be estimated assuming a conical nozzle with $\theta_{cn} = 15^\circ$ as,

$$L_n = \frac{D_e - D_t}{2 \tan \theta_{cn}}, \quad (3)$$

then $D_e = 0.9378$ cm, and let us say that a minimum mass flow of 0.5g/s is desired, thus using Eq. 3-24 from Sutton, the $D_t = 0.127$ cm for which the expansion ratio is $\epsilon = 62$. It must be remarked that this θ_{cn} is associated with a divergence correction of 0.9829, however, in this

paper, this is neglected. This provides a relatively high $C_F = 1.918$ (from Eq. 3-30 in Sutton) and $I_{sp} = 440.2$ s from Fig. 11 while maintaining small dimensions. These dimensions allow burning the gas produced by the MHD drive every hour in 6s.

A.2.2 Combustion Chamber sizing

For the sizing of the combustion chamber, the characteristic length of the combustion chamber L^* is needed. For gaseous H_2 and O_2 there is no value computed in Table 5.6. from [4] thus, the value for the liquid O_2 and gaseous H_2 is used, $L^* \in [56, 71]cm$, the high value is taken here for the characteristic length, assuming the worst situation to have margin with the dimensions.

$$V_{cc} = L^* A_t = 0.899cm^3. \quad (4)$$

Now, for the combustion chamber area, the cross-sectional area should be large enough to keep the Mach number reasonably subsonic. This ratio of D_{cc}/D_t should approximately range from 3 to 5 for small rockets. It can be found in [11] an expression that relates this ratio with the throat diameter,

$$\frac{A_{cc}}{A_t} = 8D_t^{-0.6} + 1.25, \quad (5)$$

where the D_t is in cm. This gives a $A_{cc} = 0.425 cm^2$ and $D_{cc} = 0.681 cm$ which is close to the ratio of 5 previously commented. Then, from the previously obtained V_{cc} the combustion chamber length is $L_{cc} = V_{cc}/A_{cc} = 2.46 cm$.

Overall, the system sizing is done such that the nozzle and combustion chamber system fits in a CubeSat, the total length is 4.21cm, which is feasible for CubeSat propulsion application as it would occupy less than half of 1U.

A.3 Sizing cold gas thrusters

For desaturation around the Z-axis, either thruster 3 or thruster 1 can be used. For desaturation around the Y-axis, either thruster 4 or thruster 1 can be used. Finally, for desaturation around the X-axis, either thrusters 3 and 1 simultaneously or thrusters 2 and 4 simultaneously can be used.

The torque generated by the desaturation of each axis can be expressed as a function of the geometry shown in Fig. 7(a), where $d = 0.04 m$,

$$T_z = d \cos(20^\circ) F_{3 \text{ or } 1} \quad T_y = d \cos(20^\circ) F_{4 \text{ or } 2} \quad T_x = d \cos(90 - 20^\circ) F_{3 \text{ and } 1 \text{ or } 2 \text{ and } 4}. \quad (6)$$

The required torque for desaturation in 60s is $0.521 mN \cdot m$ and using the torque expressions around the different axes in Eq. 6 the required thrust can be determined. Specifically, $F_{Y \text{ and } Z} = 13.86 mN$ is needed for desaturation around the Y and Z axes, and $F_X = 19.04 mN$ is needed for desaturation around the X axis.

Now that the required thrust is known, the mass flow rate and nozzle dimensions can be calculated using Eq. 1 and Eq. 3-24 from Sutton. However, first the specific impulse I_{sp} for the H_2 and O_2 mixture at 300K without combustion must be estimated.

The specific impulse is determined by C_F and c^* , as shown in Eq. 2. To obtain C_F for a given ϵ , Eq. 3-30 in Sutton can be used. When operating in a vacuum, p_3 is equal to zero and the relationship between p_2 and p_1 can be found using ϵ in Eq. 3-25. For a stoichiometric mixture of H_2 and O_2 at 10 bar and 300K, the specific heat ratio k is 1.401. The expression for c^* depends only on temperature, $k = 1.4$, and $MW = 12.02 g/mol$, as shown in Eq. 3-32, all of which are known. Therefore, the characteristic velocity is calculated as $c^* = 665.26 m/s$.

Iterative calculations are performed using the given data. Initially, an assumption of $\epsilon = 50$ is made, which yields a C_F value of 1.729. Subsequently, using the C_F value and the

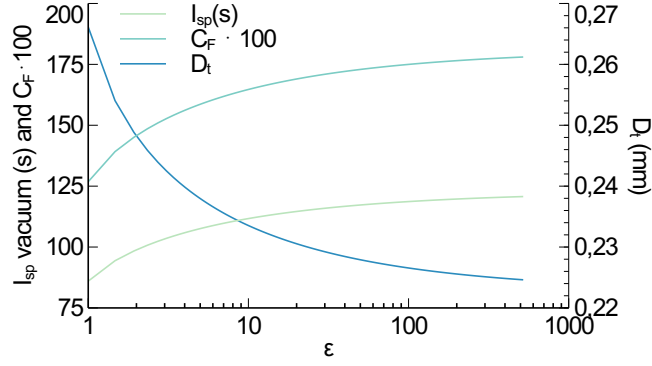


Figure 12: Variation of C_F , I_{sp} and D_t for a gaseous mixture of H_2 and O_2 at 300K and 10 bar.

known c^* , the specific impulse I_{sp} is 117.26s. It is important to note that this I_{sp} value is higher than the typical specific impulse of cold gas thrusters, which is usually around 60-70s, due to the low molecular weight of the gaseous H_2 and O_2 mixture.

Using Eq. 1, the mass flow rate $\dot{m}$ is calculated to be 0.016569g/s, and then the throat area and diameter are obtained using Eq. 2-17, resulting in $D_t = 0.09304$ mm. However, this throat area is extremely small and may be difficult to manufacture. To address this, the desaturation time requirement is changed to 10s. Repeating the calculations, a thrust of 114.24 mN is required, which is provided by a throat diameter of $D_t = 0.27$ mm, which is still small but is more feasible to be manufactured. This small throat diameter allows a higher expansion ratio ϵ since the exit diameter for this $\epsilon = 50$ is still small $D_e = 0.16$ mm. Using $\epsilon = 200$, a new C_F value of 1.7652, throat diameter of $D_t = 0.225$ mm, and exit diameter $D_e = 3.2$ mm are obtained. Assuming $\theta_{cn} = 15^\circ$, a nozzle length of L_n of 5.5mm is obtained, which are feasible dimensions for CubeSat application.

It should be noted that as the ϵ value is increased, changing the nozzle expansion ratio has a decreasing effect on the system's performance, since the C_F value asymptotically approaches a maximum value with increasing ϵ . This can be observed in Figs 3-7 and 3-8 in Sutton's book. However, these figures are not available for $k = 1.4$, so a plot of the variation of C_F , throat diameter D_t , and I_{sp} with expansion ratio is shown in Fig. 12 for the given conditions. It can be observed how the required throat diameter is reduced with the expansion ratio as the I_{sp} is increased and less mass flow rate is needed for the same thrust.

B Appendix B

B.1 Budgets

B.1.1 Mass and Volume

Subsystem	Mass(kg)	Volume(cm ³)
1.0 Science Equipment		
1.1 Laser Pointer (including PCM chasis)	1	1000
1.2 NIR reflectometer	1.2	1000
1.3 Gieger Couter	0.064	47.488
2.0 ADCS		
2.1 Sun Sensors (14)	0.028	
2.2 IMU	0.055	37.17952
2.3 Reaction Wheels	2.62	438.935625
2.4 Star Tracker	0.047	3.36
3.0 ODH		
3.1 Onboard Computer	0.176	107.136
4.0 Power		
4.1 Deployable Solar Pannels x6	0.9	
4.2 Secondary Battery	0.335	186.329948
5.0 Thermal		
5.1 Passive Louvers	0.965	218.3
5.2 AZ-93 Thermal Paint	0.126	15.65
6.0 Telecom		
6.1 Antenna	0.09	
6.2 Receiver/Transmitter	0.075	129.6
7.0 Propulsion		
Propulsion System	2	
Propellant	0.4	120.21
8.0 Structure		
Internal Structures	0.25	100
6U Frame	0.9	7695.3
Total Mass / Volume	11.231	3433.748693

Figure 13: Mass and Volume Budgets

B.1.2 Power

Subsystems	Power Modes, (We)						
	Cruise Phase Modes			Science Phase Modes			End of Life
	Mode 1 Launch	Mode 2 Cruise	Mode 3 NRHO Insertion	Mode 4 Nominal	Mode 5 Data Collection	Mode 6 Data Transmission	
Orbiter Stage (NRHO)	20.95	53.321	111.051	24.051	126.98	24.051	22.021
1.0 Instruments	0.14	0.14	0.14	0.14	101.14	0.14	0.14
1.1 Laser Pointer	0	0	0	0	100	0	0
1.2 NIR reflectometer	0	0	0	0	1	0	0
1.2 Geiger Counter (2)	0.14	0.14	0.14	0.14	0.14	0.14	0.14
2.0 ADCS	20.21	18.481	18.481	18.481	20.21	18.481	20.481
2.1 Sun Sensors (14)	0.21	0.21	0.21	0.21	0.21	0.21	0.21
2.2 IMU	2	0	0	0	2	0	2
2.3 Reaction Wheels (3)	18	18	18	18	18	18	18
2.4 Star Tracker	0	0.271	0.271	0.271	0	0.271	0.271
2.5 Thrusters	0	0	0	0	0	0	0
3.0 ODH	0.6	0.4	0.4	0.4	0.6	0.4	0.4
3.1 Onboard Computer	0.6	0.4	0.4	0.4	0.6	0.4	0.4
4.0 Power	0	0	0	0	0	0	0
4.1 Deployable Solar Panels							
4.2 Secondary Battery							
5.0 Thermal	0	0	0	0	0	0	0
5.1 Radiator	0	0	0	0	0	0	0
6.0 Telecom	0	6.3	4.03	4.03	4.03	4.03	0
6.1 Antenna	0	2.3	0.03	0.03	0.03	0.03	0
6.2 Receiver/Transmitter	0	4	4	4	4	4	0
7.0 Propulsion	0	28	88	1	1	1	1
7.1 Valves and pressure sensor	0	1	1	1	1	1	1
7.2 Electrolysis	0	0	27	0	0	0	0
7.3 Spark plug	0	27	60	0	0	0	0
Total Power / Mode	20.95	53.321	111.051	24.051	126.98	24.051	22.021

Figure 14: Power Budget

Degradation rate and power at end of life were calculated using Equations 7 and 8 below taken from SMAD [12].

$$\eta_{degradation} = (1 - \zeta)^{\tau} \quad (7)$$

$$PEOL = PBOL \times \eta_{degradation} \quad (8)$$

B.1.3 Cost

	Phase A		Phase B		Phase C		Phase D		Phase E		Phase F	
	FY2023	Total	FY2024	Total	FY2025	Total	FY2026	Total	FY2027	Total	FY2027	Total
project management	\$8,400.00	\$8,400.00	\$8,400.00	\$8,400.00	\$16,800.00	\$17,814.72	\$58,800.00	\$62,351.52	\$121,151.52	\$67,200.00	\$71,258.88	\$138,458.88
systems engineering	\$11,200.00	\$11,200.00	\$11,200.00	\$11,200.00	\$22,400.00	\$23,752.96	\$78,400.00	\$83,135.36	\$161,535.36	\$89,600.00	\$95,011.84	\$184,611.84
safety and mission assurance	\$3,920.00	\$3,920.00	\$3,920.00	\$3,920.00	\$7,840.00	\$8,313.54	\$27,440.00	\$29,097.38	\$56,537.38	\$31,360.00	\$33,254.14	\$64,614.14
science/technology	\$5,600.00	\$5,600.00	\$5,600.00	\$5,600.00	\$11,200.00	\$11,876.48	\$39,200.00	\$41,567.68	\$80,767.68	\$44,800.00	\$47,505.92	\$92,305.92
payloads						0	0	0	0	0	0	0
Sensors						0	0	5500	5832.2	11332.2	0	0
spacecraft						0	0	0	0	0	0	0
Propulsion						0	0	10000	10604	20604	0	0
Structure					7500	7953	15453	0	0	0	0	0
ADCS						0	0	0	0	0	0	0
-Thruster						0	0	10000	10604	20604	0	0
-Reaction wheels						0	0	28000	29691.2	57691.2	0	0
-star tracker						0	0	17600	18663.04	36263.04	0	0
-IMU							0	4000	4241.6	8241.6		
-sun sensors						0	0	4200	4453.68	8653.68	0	0
solar panels								32191	34135.3364	66326.3364		
Power						0	0	3000	3181.2	6181.2	0	0
Comms						0	0	5000	5302	10302	11584.06494	13812.43161
Thermal						0	0	0	0	0	25396.49566	16489.45766
-Passive Louvers						0	0	15.5	16.4362	31.9362	0	0
-AZ93 Thermal Paint						0	0	100	106.04	206.04		
Mission Operations	\$15,120.00	\$15,120.00	\$15,120.00	\$15,120.00	\$30,240.00	\$32,066.50	\$105,840.00	\$112,232.74	\$218,072.74	\$120,960.00	\$128,265.98	\$249,225.98
Launch Vehicle/Services						0	0	270000	286308	556308	270000	286308
Ground Systems	\$15,120.00	\$15,120.00	\$15,120.00	\$15,120.00	\$30,240.00	\$32,066.50	\$105,840.00	\$112,232.74	\$218,072.74	\$120,960.00	\$128,265.98	\$249,225.98
Systems Integration & testing						\$0.00	\$0.00	\$129,360.00	\$137,173.34	\$266,533.34	\$304,609.54	\$611,142.88
Education and public outreach	\$1,120.00	\$1,120.00	\$1,120.00	\$1,120.00	\$2,240.00	\$2,375.30	\$7,840.00	\$8,313.54	\$16,153.54	\$8,960.00	\$9,501.18	\$18,461.18
TOTAL	\$4,601,252.43											

Figure 15: Cost Budget